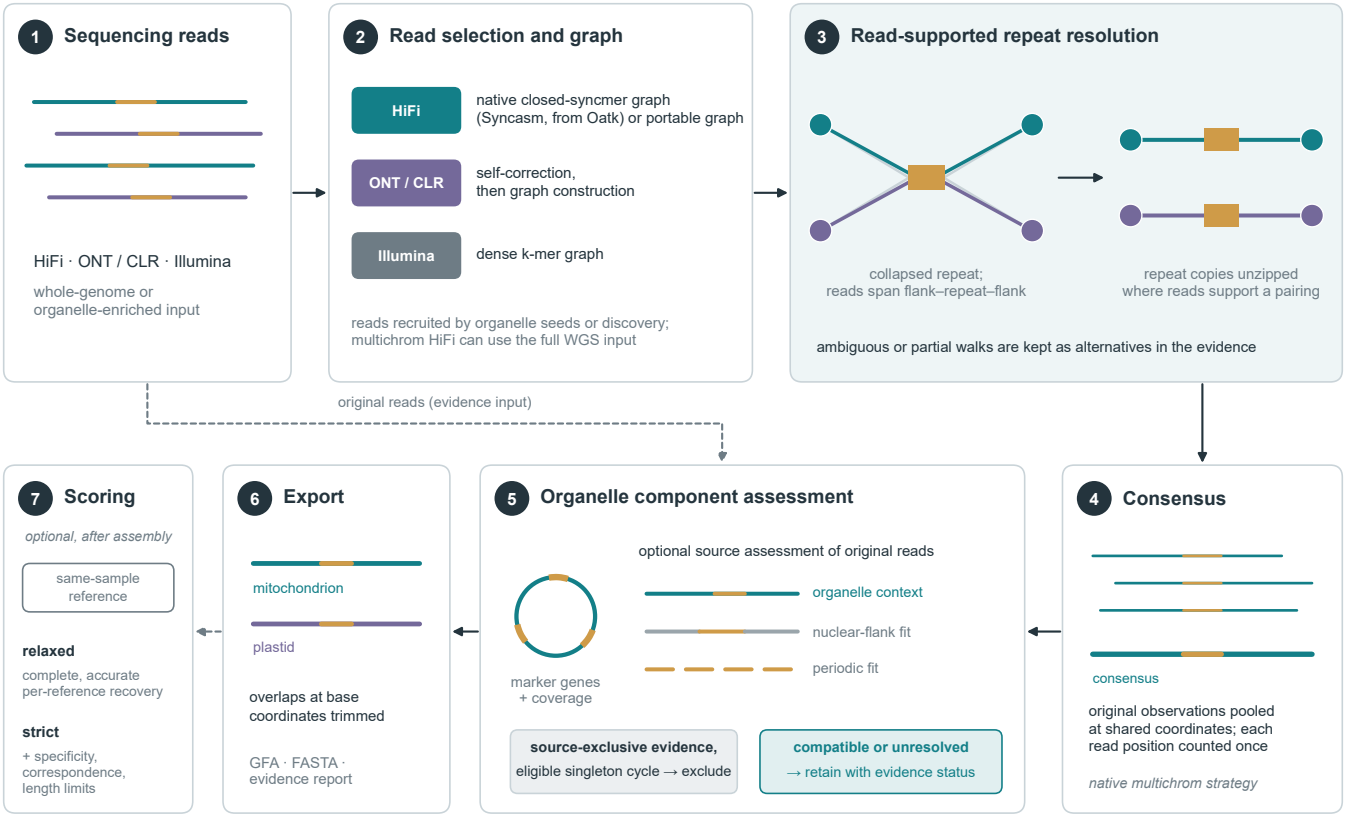

1 Sequencing reads
HiFi · ONT / CLR · Illumina
whole-genome or organelle-enriched input
2 Read selection and graph
HiFi
native closed-syncmer graph (Syncasm, from Oatk) or portable graph
ONT / CLR
self-correction, then graph construction
Illumina
dense k-mer graph
reads recruited by organelle seeds or discovery; multichrom HiFi can use the full WGS input
3 Read-supported repeat resolution
collapsed repeat; reads span flank–repeat–flank
repeat copies unzipped where reads support a pairing
ambiguous or partial walks are kept as alternatives in the evidence
original reads (evidence input)
4 Consensus
consensus
original observations pooled at shared coordinates; each read position counted once
native multichrom strategy
5 Organelle component assessment
optional source assessment of original reads
organelle context
nuclear-flank fit
periodic fit
marker genes + coverage
source-exclusive evidence, eligible singleton cycle → exclude
compatible or unresolved → retain with evidence status
6 Export
mitochondrion
plastid
overlaps at base coordinates trimmed
GFA · FASTA · evidence report
7 Scoring
optional, after assembly
same-sample reference
relaxed
complete, accurate per-reference recovery
strict
+ specificity, correspondence, length limits
Graph cycles and FASTA records do not identify physical molecule number or topology; geometry is schematic.
distinguishable sequence contexts
shared or repeated sequence
processing
evidence input or optional step